

packages/graph-core/src/__tests__/fingerprint.test.ts

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Overview

A new test file `packages/graph-core/src/__tests__/fingerprint.test.ts` (lines R1-R427) has been added.

It imports the fingerprint utilities (`contentHash` , `extractFileFingerprint` , `compareFingerprints` , `analyzeChanges`) and the `StructuralAnalysis` type from `../types.js` .

The file mocks `node:fs` (lines R12-R18) to provide `readFileSync` and `existsSync` stubs, and clears these mocks before each test (line R23).

All tests run under `vitest` (imported at R1).

Key changes

- **Imports** – `vitest` helpers, `StructuralAnalysis` , and the four fingerprint functions (R1-R10).
- **Mocking** – `vi.mock("node:fs")` replaces the real file-system functions (R12-R16).
- **Content-hash tests** – verify deterministic SHA-256 output and distinct hashes for different inputs (R27-R34).
- **Fingerprint extraction tests** – cover functions, classes, imports, exports, content hash, and line counts (R40-R126).
- **Fingerprint comparison tests** – exercise all change levels (`NONE` , `COSMETIC` , `STRUCTURAL`) and edge cases such as missing structural analysis (R128-R330).
- **Analyze-changes tests** – validate classification of new, deleted, unchanged, and unknown files, and confirm that input arrays are not mutated (R332-R427).
- **Mock usage** – `mockedReadFileSync` and `mockedExistsSync` are cast with `vi.mocked` (R20-R21).
- **Registry and store** – a mock registry (`analyzeFile`) and an existing fingerprint store are used in the `analyzeChanges` tests (R333-R361).

Impact

- **Coverage** – the test suite exercises every exported fingerprinting function and the `analyzeChanges` workflow, providing early detection of regressions.
- **Maintainability** – changes to fingerprint logic will be caught by the focused test blocks.
- **Observability** – failures surface immediately in CI with clear diagnostics on which comparison rule failed.
- **Performance** – unknown from the available diff/scan evidence.

Risks & follow-ups

- **Mock reliability** – ensure `vi.mocked` correctly casts the stubs; run tests to confirm no TypeScript errors.

- **Snapshot drift** – if the hash algorithm or fingerprint format changes, snapshots may need updating (`vitest --update`).
- **Edge-case coverage** – verify that `analyzeChanges` handles files lacking structural analysis; targeted tests already exist (R304-R330).
- **CI integration** – confirm the new test file is included in the test matrix and that the `vitest` command is updated accordingly.